

Numerical Analysis Exercise 3

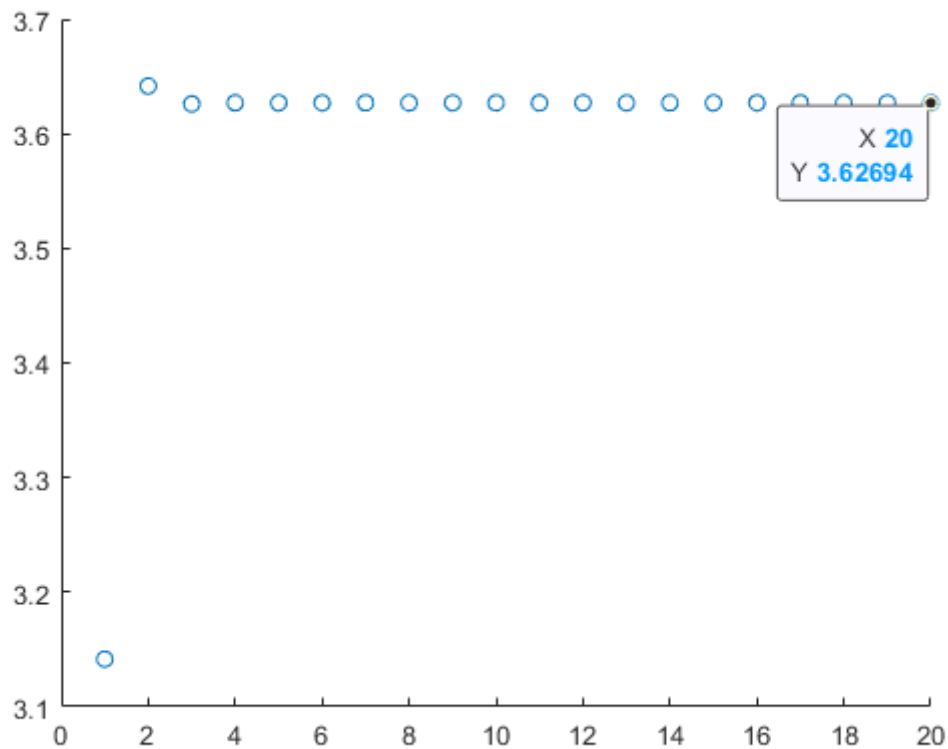
Problem1

(1). Use Theorem 2.2 to show that $g(x) = \pi + 0.5 \sin(x/2)$ has a unique fixed point on the interval $[0, 2\pi]$;

Solution For $x \in [0, 2\pi]$, the absolute minimum of g occurs at $x = 0$ and $g(0) = \pi - 0.5$, and the absolute maximum of g occurs at $x = \pi$ and $g(\pi) = \pi + 0.5$. So $g(x) \in [\pi - 0.5, \pi + 0.5]$ for all $x \in [0, 2\pi]$. In addition, g' is continuous and $|g'(x)| = \frac{1}{4} < 1$ for $x \in [0, 2\pi]$. Therefore, g satisfies all hypotheses of Theorem 2.2 in $[0, 2\pi]$

```
1 p=zeros(1,20);
2
3 p(1)=pi;%初值
4
5 for i=1:19
6     p(i+1)=pi+0.5*sin(p(i)/2);
7 end
8 p
```

```
1 p = 1×20
2     3.1416     3.6416     3.6260     3.6270     3.6269     3.6269     3.6269
   3.6269     3.6269     3.6269     3.6269     3.6269     3.6269     3.6269
   3.6269     3.6269     3.6269     3.6269     3.6269     3.6269
```



This figure is the fixed points, it is easy to see that the points are convergent to 3.6294

(2). Use the fixed point iteration to find the approximation to the fixed point that is accurate to within 10^{-2} .

Solution

```

1 k=1/4;%g'(x)<=1/4;
2 Error=zeros(1,20);
3
4 for n=1:20
5     Error(1,n)=k^n/(1-k)*abs(p(2)-p(1));
6 end
7
8 error=Error(Error<=1E-2)

```

```

1 error = 0.00260416666666667 0.000651041666666667
0.0001627604166666667 4.06901041666667e-05 1.01725260416667e-05
2.54313151041667e-06 6.35782877604167e-07 1.58945719401042e-07
3.97364298502604e-08 9.93410746256510e-09 2.48352686564128e-09
6.20881716410319e-10 1.55220429102580e-10 3.88051072756449e-11
9.70127681891123e-12 2.42531920472781e-12 6.06329801181952e-13

```

Problem 2

Use Newton method to get the approximations to the following numbers respectively (accurate to within 10^{-4}) and show your Matlab or Python code.

Solution $f(x) = x^2 - a, a \in [2, 13]$ First, easy to know that $f(x) \in C^2$. then, because x is the Prime Numbers in $[2, 13]$, so $f'(x) \neq 0$, therefor, we can use the *Newton-Raphson method*

```
1  clc, clear all
2
3  a=input("Please input the number:")
4  newtonsqrt(a,1e-4)
5
6  function approximation = newtonsqrt(a,accuracy)
7  q=input("Please input the initial number:"); %设置初值
8  p=0.5*(q+a/q);
9  while (abs(p-q)>=accuracy)
10     q=p; %后一个值迭代到当前值
11     p=0.5*(q+a/q);
12 end
13 p
14 end
```

```
1  a = 5
2  p = 2.2361
```